



DC Motor Control and Data Acquisition System (STM32 + Qt)

STM32

CUBEIDE

Qt QT

C++

HARDWARE

IN-THE-LOOP

This repository contains the firmware (C) and graphical user interface (C++) developed for the real-time control and monitoring of a Direct Current (DC) Motor. The project uses an STM32 microcontroller interacting with a desktop application via Serial (UART) communication, allowing for precise analyses of the motor's dynamics (τ mechanical, current, speed ramp).

☆☆ Main Features (Infrastructure for *Asservissement*)

- **Signal Acquisition for Closed-Loop Control:**
 - **Dynamic Speed (A/B Channels):** Read via a hardware timer in quadrature at 100 Hz, ensuring the high resolution and stability required for future PID control.
 - **Absolute Validation (I Channel):** Read via external interrupt (EXTI) acting as a "source of truth" to confirm encoder pulses and ensure no accumulated error in the machine.
 - **Motor Current:** Read via ADC with the implementation of a Low-Pass Filter (smoothing) for torque monitoring and electrical safety.
- **Real-Time Telemetry:** Desktop interface developed in Qt with continuous plotting of the motor's response graphs (RPM and Current) using the **QCustomPlot** library.
- **Safety and Actuation:** Power drive via a 20 kHz PWM signal (highly efficient and inaudible) integrated with a hardware-isolated Emergency Brake system for workbench protection.

🔧 Hardware Architecture

- **Controller Board:** NUCLEO-F411RE (ARM Cortex-M4)
- **Power Driver:** H-Bridge / *Hacheur* compatible with PWM signals.
- **Actuator/Sensors:** DC Motor coupled with an Optical/Magnetic Encoder (A, B, and I channels).
- **Current Sensor:** Analog module read via ADC. (*Note: A hardware voltage divider was used to protect the STM32's 3.3V ADC against 4V/5V spikes.*)

Main Pinout (STM32)

Physical Pin	Microcontroller Function	Destination
PA8 (D7)	TIM1_CH1 (Advanced PWM)	Power Signal (Driver)
PA6 / PA9	GPIO Output	Direction of Rotation / Emergency Brake
PB4 / PB5	TIM3 (Encoder Mode TI12)	Encoder Channels A and B
PB10 (D6)	EXTI (External Interrupt)	Encoder Channel I (Index)

Physical Pin	Microcontroller Function	Destination
PA0 (A0)	ADC1 Channel 0	Current Sensor

Développé dans le cadre du module d'Asservissement / Systèmes Commandés.

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